

WEEK-2

```
1 • DROP DATABASE IF EXISTS bookflow_db;
2 • CREATE DATABASE bookflow_db;
3 • USE bookflow_db;
4
5 • CREATE TABLE Books (
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK (published_year < 2027)
10 );
11
12 • INSERT INTO Books (book_id, title, isbn, published_year)
13 VALUES
14 (1, 'The Great Gatsby', '9780743273565', 1925),
15 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
16 (3, '1984', '9780451524935', 1949);
17
18 • SELECT * FROM Books;
19
20 • CREATE TABLE Members (
21     member_id INT PRIMARY KEY,
22     full_name VARCHAR(100),
23     email VARCHAR(100) UNIQUE
```

Output

#	Time	Action	Message
✓ 1	18:22:12	DROP DATABASE IF EXISTS bookflow_db	1 row(s) affected
✓ 2	18:22:14	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	18:22:16	USE bookflow_db	0 row(s) affected
✓ 4	18:22:18	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20)...	0 row(s) affected
✓ 5	18:22:21	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925). ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

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```
3 • USE bookflow_db;
4
5 • CREATE TABLE Books (
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK (published_year < 2027)
10 );
11
12 • INSERT INTO Books (book_id, title, isbn, published_year)
13 VALUES
14 (1, 'The Great Gatsby', '9780743273565', 1925),
15 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
16 (3, '1984', '9780451524935', 1949);
17
18 • SELECT * FROM Books;
```

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Books 1

Output

#	Time	Action	Message
✓ 1	18:22:12	DROP DATABASE IF EXISTS bookflow_db	1 row(s) affected
✓ 2	18:22:14	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	18:22:16	USE bookflow_db	0 row(s) affected
✓ 4	18:22:18	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
✓ 5	18:22:21	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 19...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 6	18:22:37	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

WEEK-2

```

30 (103, 'Michael Brown', 'michael.brown@email.com');
31
32 • SELECT * FROM Members;
33
34 • CREATE TABLE Loans (
35     loan_id INT PRIMARY KEY,
36     member_id INT,
37     book_id INT,
38     loan_date DATE,
39     FOREIGN KEY (member_id) REFERENCES Members(member_id),
40     FOREIGN KEY (book_id) REFERENCES Books(book_id)
41 );
42
43 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
44 VALUES
45 (1, 101, 1, '2025-01-05'),

```

Result Grid

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

Members 2

Output

Action Output

#	Time	Action	Message
2	18:22:14	CREATE DATABASE bookflow_db	1 row(s) affected
3	18:22:16	USE bookflow_db	0 row(s) affected
4	18:22:18	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
5	18:22:21	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	18:22:37	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
7	18:24:27	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR...	0 row(s) affected
8	18:24:30	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (1...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	18:24:43	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

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```

48 (4, 101, 2, '2025-02-01'),
49 (5, 102, 1, '2025-02-05'),
50 (6, 103, 2, '2025-02-12'),
51 (7, 101, 3, '2025-03-01'),
52 (8, 102, 3, '2025-03-07'),
53 (9, 103, 1, '2025-03-15'),
54 (10, 101, 1, '2025-04-01'),
55
56 • SELECT * FROM Loans;
57
58 • LEFT

```

Result Grid

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

Loans 3

Output

Action Output

#	Time	Action	Message
5	18:22:21	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	18:22:37	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
7	18:24:27	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR...	0 row(s) affected
8	18:24:30	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (1...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	18:24:43	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
10	18:24:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
11	18:25:02	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	18:25:05	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned

Limit to 1000 rows

```

57
58 • SELECT
59     m.full_name AS Member_Name,
60     b.title AS Book_Title
61 FROM Loans l
62 INNER JOIN Members m
63 ON l.member_id = m.member_id
64 INNER JOIN Books b
65 ON l.book_id = b.book_id;
66
67 • SELECT
68     published_year,
69     COUNT(book_id) AS Total_Books
70 FROM Books
71 GROUP BY published_year
72 ORDER BY published_year;

```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 5 x

Output

Action Output

#	Time	Action	Message
7	18:24:27	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100));	0 row(s) affected
8	18:24:30	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	18:24:43	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
10	18:24:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATETIME);	0 row(s) affected
11	18:25:02	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'); (2, 102, 2, '2025-01-05');	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
12	18:25:05	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
13	18:25:48	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id INNER JOIN Books b ON l.book_id = b.book_id;	10 row(s) returned
14	18:26:09	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year;	3 row(s) returned

Limit to 1000 rows

```

85 VALUES
86 (4, 'Animal Farm', '9780451526342', 1945);
87
88 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
89 VALUES
90 (1, 4, 'Raj Kumar', CURDATE());
91
92 • COMMIT;
93
94 • CREATE INDEX idx_books_isbn
95 ON Books(isbn);
96
97 • SELECT *
98 FROM Books
99 WHERE isbn = '9780451524935';
100

```

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949

Books 6 x

Output

Action Output

#	Time	Action	Message
14	18:26:09	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year;	3 row(s) returned
15	18:26:40	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATETIME);	0 row(s) affected
16	18:26:43	START TRANSACTION	0 row(s) affected
17	18:26:46	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945);	1 row(s) affected
18	18:26:51	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDATE());	1 row(s) affected
19	18:26:54	COMMIT	0 row(s) affected
20	18:26:57	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
21	18:27:03	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned

```

90 (1, 4, 'Raj Kumar', CURDATE());
91
92 • COMMIT;
93
94 • CREATE INDEX idx_books_isbn
95 ON Books(isbn);
96
97 • SELECT *
98 FROM Books
99 WHERE isbn = '9780451524935';
100
101 • SELECT * FROM Donation_History;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	donation_id	book_id	donor_name	donation_date
▶	1	4	Raj Kumar	2026-08-05
•	NULL	NULL	NULL	NULL

Donation_History7 x

Output			
Action Output			
#	Time	Action	Message
✓ 15	18:26:40	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VAR...	0 row(s) affected
✓ 16	18:26:43	START TRANSACTION	0 row(s) affected
✓ 17	18:26:46	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Fam', '9780451526342', 1945)	1 row(s) affected
✓ 18	18:26:51	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kum...	1 row(s) affected
✓ 19	18:26:54	COMMIT	0 row(s) affected
✓ 20	18:26:57	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 21	18:27:03	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned
✓ 22	18:27:45	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned